VICTORY 2027

LIGHT

LECTURE-02 PHYSICS

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Topics To Be Covered

1. REVISION
2. REFRACTION OF LIGHT THROUGH A GLASS PRISM
3. REAL AND APPARENT DEPTH
4. CRITICAL ANGLE AND TOTAL INTERNAL REFLECTION


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REVISION

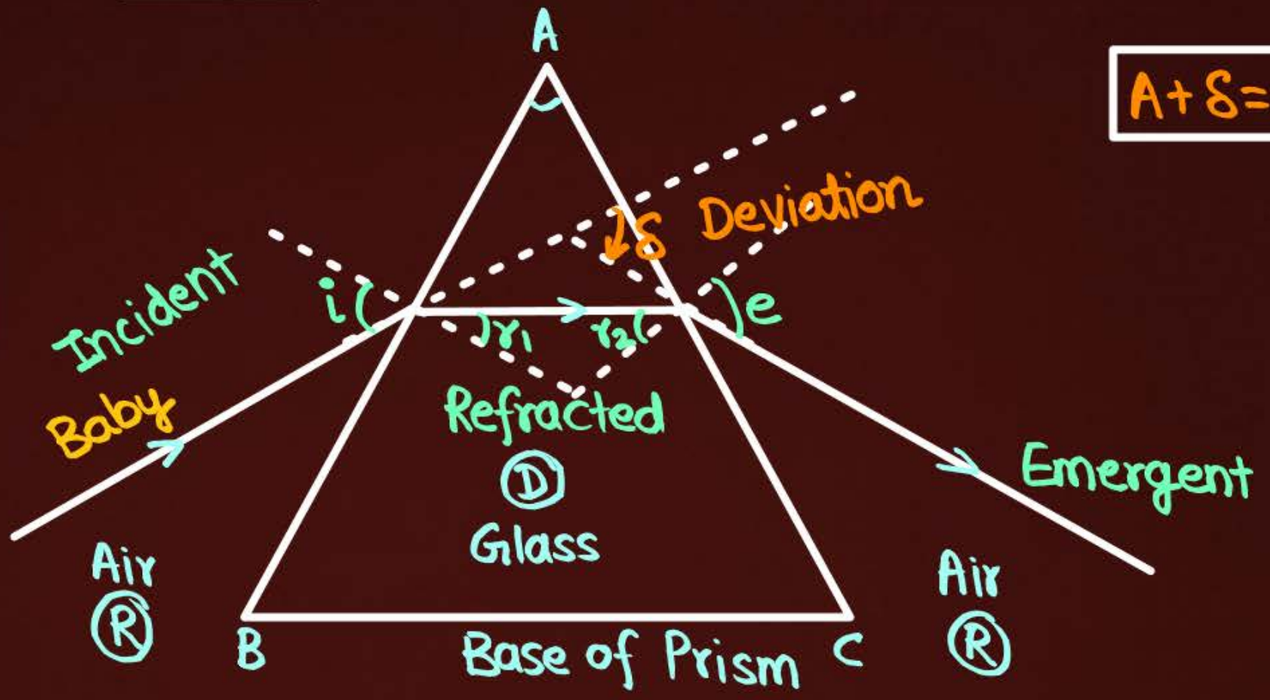
Refraction of Light

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

$f \rightarrow$ constant

$$\lambda_2 = \frac{\lambda_1}{{}_1\mu_2}$$

Prism



$A + \delta = i + e$

SSS Baby Ko Base Pasand Hai.

Formula

Refractive Index of Prism

$$\mu = \frac{\sin\frac{A+\delta_m}{2}}{\sin\frac{A}{2}}$$

A → Angle of Prism

δ_m → Minimum Deviation.

For Small Prism ($A < 10^\circ$)

$$\delta_m = (\mu - 1)A$$

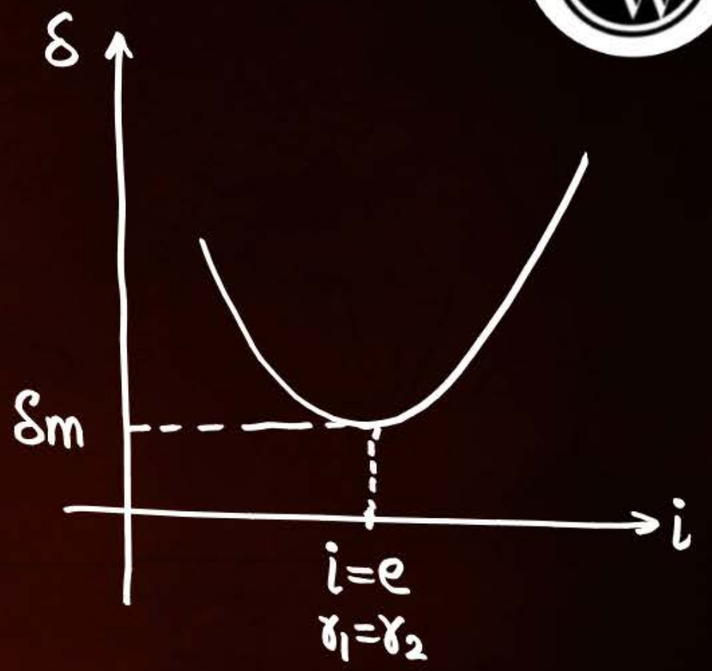
Deviation $\propto$ Refractive Index

$$\propto \frac{1}{\text{Speed}}$$

$$\propto \frac{1}{\text{Wavelength}}$$

$\delta\uparrow \ \mu\uparrow \ v\downarrow \ \lambda\downarrow$

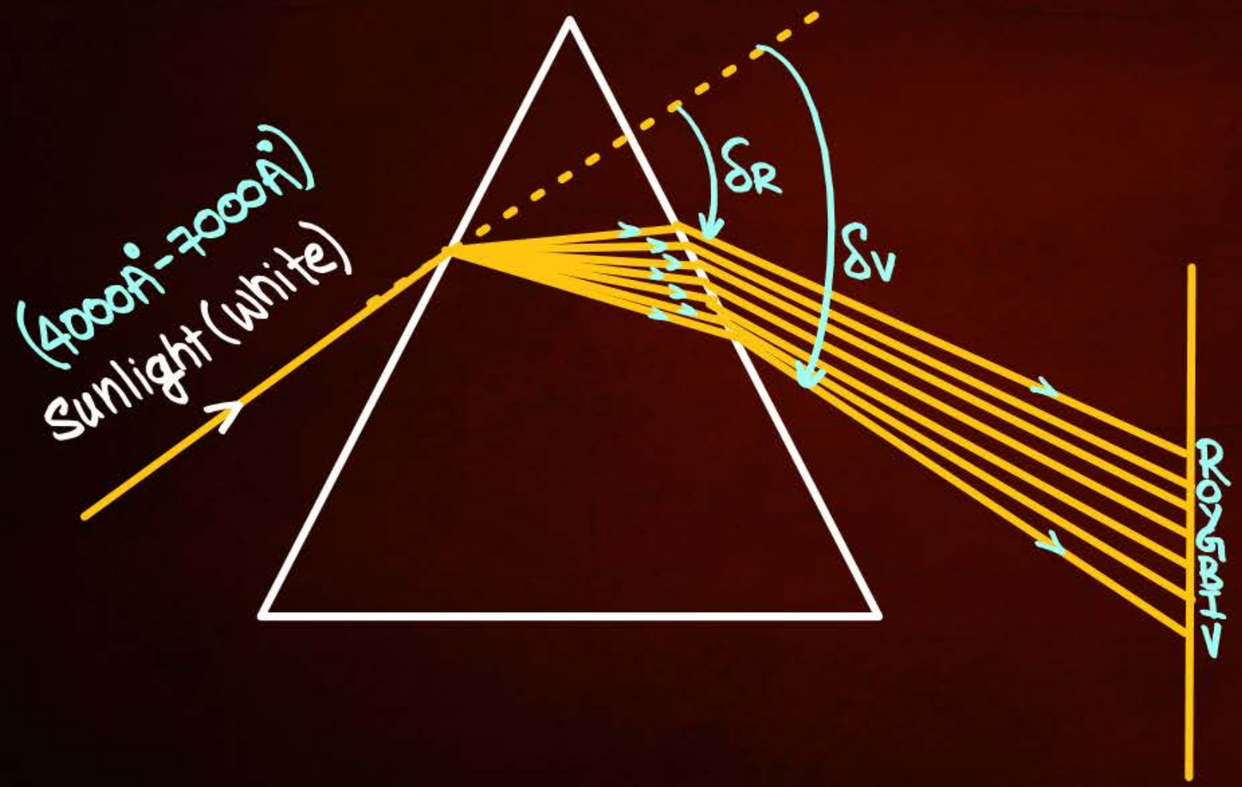
Observation:



Different Colours have different wavelength. They have different speeds as well as refractive index.

Imp

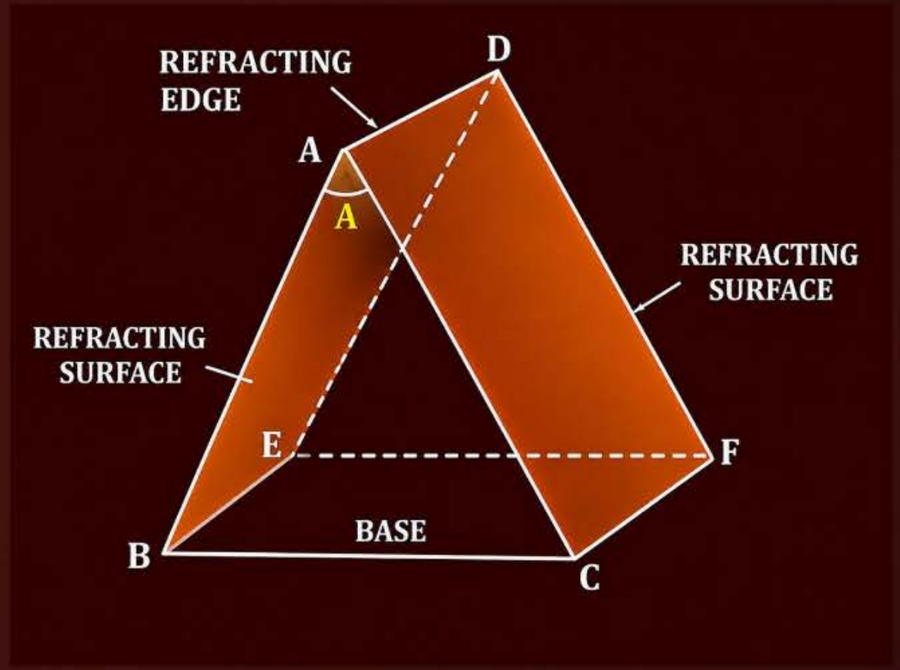
δ_{max} → Violet	δ_{min} → Red
μ_{max} → Violet	μ_{min} → Red
v_{min} → "	v_{max} → "
λ_{min} → "	λ_{max} → "



Visible Spectrum of Sunlight

PRISM

A prism is a transparent refracting medium bounded by five plane surfaces with a triangular cross section.



REFRACTION OF LIGHT THROUGH A GLASS PRISM

Note:

- In a prism, the refraction of light occurs at two inclined faces, so the emergent ray is not parallel to the incident ray, but it is deviated towards the base of prism.
- On the other hand, in a parallel sided glass slab, the refraction of light occurs at two parallel faces, so the emergent ray is parallel to the incident ray with a lateral shift.

FACTORS AFFECTING IN ANGLE OF DEVIATION

(1) the angle of incidence (i),

(2) the material of prism (i.e., on refractive index μ),

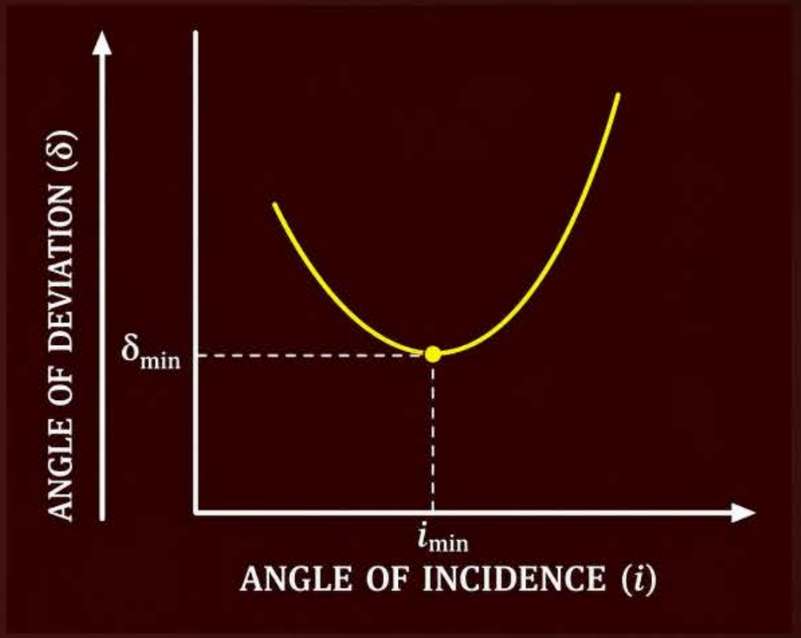
(3) the angle of prism (A), and

(4) the colour or wavelength (λ) of light used.

DEPENDENCE OF ANGLE OF DEVIATION ON THE ANGLE OF INCIDENCE ; i–δ GRAPH

It is experimentally observed that as the angle of incidence increases, the angle of deviation first decreases, reaches to a minimum value and then begins to increase.

It is found that the angle of deviation becomes minimum ($\boldsymbol{\delta_{min}}$) when the angle of incidence is equal to the angle of emergence i.e., when $\boldsymbol{i_1 = i_2}$ or when $\boldsymbol{r_1 = r_2}$.



DEPENDENCE OF ANGLE OF DEVIATION ON THE ANGLE OF INCIDENCE ; i–δ GRAPH

It is found that for a given angle of incidence, the prism with a higher refractive index produces a greater deviation than the prism which has a lower refractive index. $\delta = (\mu - 1)A \Rightarrow \delta \propto \mu$

- A flint glass prism produces more deviation than a crown glass prism of same refracting angle since

$$\mu_{flint} > \mu_{crown}$$

$$\delta_{flint} > \delta_{crown}$$

DEPENDENCE OF ANGLE OF DEVIATION ON THE ANGLE OF PRISM

It is found that the angle of deviation (δ) increases with the increase in the angle of prism (A).

For small angle of prism A, angle of deviation

$$\boldsymbol{\delta = (\mu - 1)A}$$

DEPENDENCE OF ANGLE OF DEVIATION ON THE COLOUR (OR WAVELENGTH) OF LIGHT

The refractive index of a given transparent medium is different for the light of different colours. It decreases with the increase in the wavelength of light.

- Thus the refractive index of the material of a prism is maximum for the violet light and minimum for the red light.
- Consequently, a given prism deviates the violet light most and the red light least (i.e., $\delta_{violet} > \delta_{red}$ since $\mu_{violet} > \mu_{red}$).

$$\delta \propto \mu \propto \frac{1}{v} \propto \frac{1}{\lambda}$$

QUESTIONS

In refraction of light through a prism, the light ray:

1. suffers refraction only at one face of the prism
2. emerges out from the prism in a direction parallel to the incident ray
3. bends at both the surfaces of prism towards its base
4. bends at both the surfaces of prism opposite to its base.

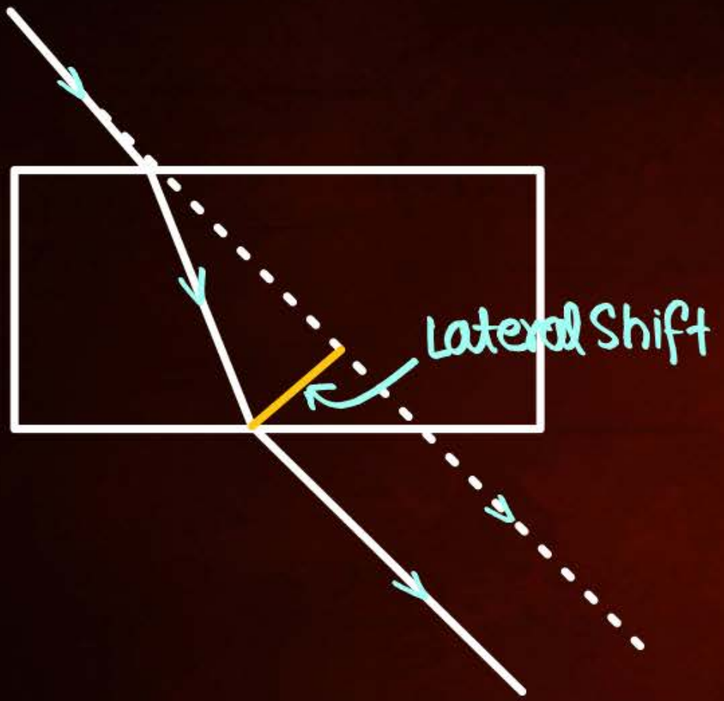
QUESTIONS

A ray of light suffers refraction through an equilateral prism. The deviation produced by the prism does not depend on the:

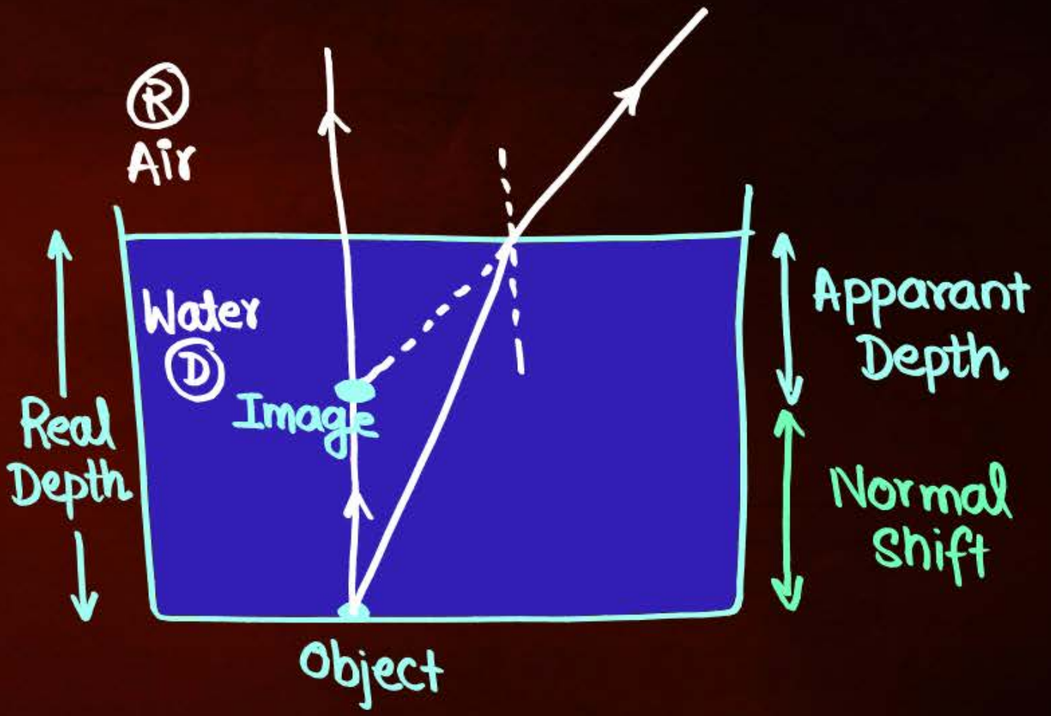
$A+\delta=i+e$

1. angle of incidence
2. colour of light
3. material of prism
4. size of prism.

Normal Shift



$$A.D. = \frac{R.D.}{a\mu_w}$$



$$N.S. = RD - AD$$

$$= RD - \frac{RD}{a\mu_w}$$

$$NS = RD\left(1 - \frac{1}{a\mu_w}\right)$$

REAL AND APPARENT DEPTH

An object placed in a denser medium when viewed from a rarer medium, appears to be at a depth lesser than its real depth. This is because of refraction of light.

REAL AND APPARENT DEPTH

The shift by which the object appears to be raised, depends on:

(1) the refractive index of the medium,

(2) the thickness of the denser medium, and

(3) the colour (or wavelength) of incident light

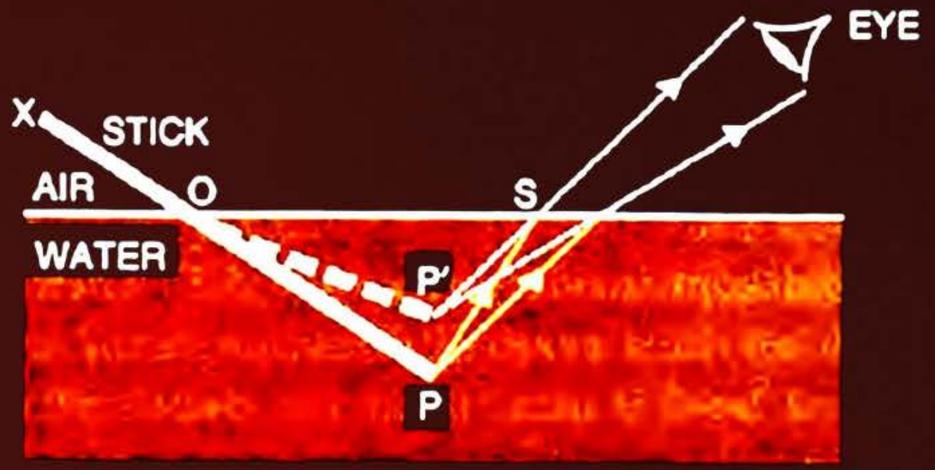
(1) **Dependence of shift on the refractive index:** Higher the refraction index of the medium, more is the shift.

(2) **Dependence of shift on the thickness of medium:** Thicker the medium, more is the shift.

(3) **Dependence on wavelength (or colour) of light:** The shift decreases with the increase in the wavelength of light used. Since $\mu_V > \mu_R$, therefore the shift is more for the violet light than for the red light in a given medium.

APPARENT BENDING OF A STICK UNDER WATER

The portion OP of the stick (or pencil) under water when seen from air appears to be shortened and raised up as OP'. This is due to refraction of light from water to air.

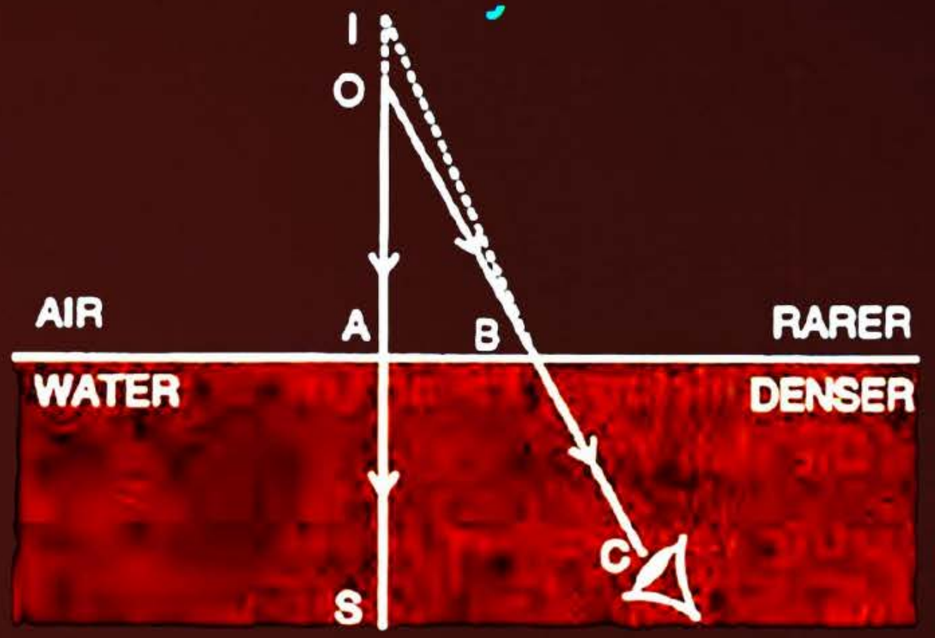


- The rays of light coming from tip P of the stick (or pencil) when pass from water to air, bend away from the normal and appear to be coming from a point P' which is the virtual image of the point P . The same is true for every point of the stick (or pencil) inside water from P to O .
- Thus, the part PO of the stick (or pencil) appears to be $P'O$, i.e., the immersed part of the stick appears to be raised and therefore bent at the point O on the surface of water and the stick (or pencil) XOP appears as XOP'.

APPARENT BENDING OF A STICK UNDER WATER

NOTE:

An object placed in a rarer medium when viewed from a denser medium appears to be at a greater distance than its real distance. In Fig., an object O placed in air when viewed from water appears to be at I which is higher than the object O .



SOME CONSEQUENCES OF THE REFRACTION OF LIGHT

(i) A star appears twinkling in the sky due to change in refractive index of air with temperature.

(ii) The sun is seen a few minutes before it rises above the horizon in the morning while in the evening few minutes longer after it sets.

(iii) A coin kept in a vessel and not visible when seen from just below the edge of the vessel, can be viewed from the same position when water is poured into the vessel.

QUESTIONS

The apparent depth of a liquid in a vessel is 15 cm , when its real depth is 20 cm . Find the refractive index of the liquid.

1. 1.2
2. 1.33
3. 1.5
4. 1.8

$$AD = \frac{RD}{\mu}$$

$$15 = \frac{20}{\mu}$$

$$\mu = \frac{4}{3}$$

QUESTIONS

A coin kept inside water ($\mu = 4/3$) when viewed from air in a vertical direction, appears to be raised by 2.0 mm. Find the depth of coin in water.

1. 2 mm
2. 4 mm
3. 6 mm
4. 8 mm

$$NS = RD\left(1 - \frac{1}{\mu}\right)$$

$$2 = RD\left(1 - \frac{1}{\left(\frac{4}{3}\right)}\right)$$

$$2 = RD\left(\frac{1}{4}\right)$$

$$RD = 8 \text{ mm}$$

QUESTIONS

A small air bubble in a glass block when seen from above appears to be raised because of:

1. refraction of light
2. reflection of light
3. reflection and refraction of light
4. none of the above.

QUESTIONS

An object in a denser medium when viewed from a rarer medium appears to be raised. The shift is maximum for:

$$NS = RD\left(1 - \frac{1}{\mu}\right)$$

$\mu \uparrow \quad NS \uparrow$

1. red light
2. violet light
3. yellow light
4. green light.

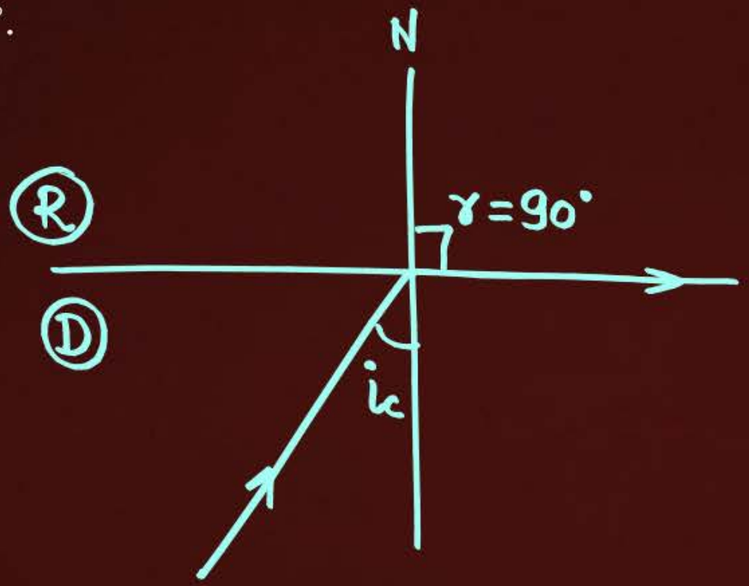
QUESTIONS

When a ray of light enters from a denser medium to a rarer medium, then: [ICSE SQP 2024]

1. the light ray bends towards the normal
2. the speed of light increases
3. the angle of incidence is greater than the angle of refraction.
4. its wavelength decreases.

CRITICAL ANGLE

Critical angle is the angle of incidence in the denser medium corresponding to which the angle of refraction in the rarer medium is $90°$.



$$\sin i_c = \frac{1}{{}_rμ_d}$$

$$\frac{\sin i_c}{\sin 90^\circ} = {}_Dμ_R$$

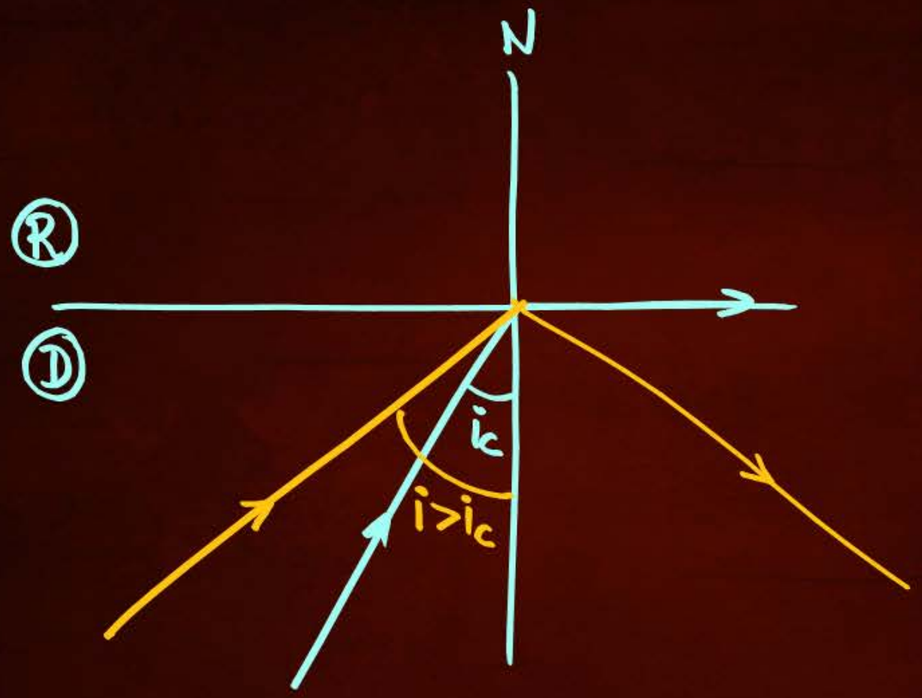
$$\sin i_c = {}_Dμ_R$$

FACTORS AFFECTING THE CRITICAL ANGLE

(1) **DEPENDENCE ON THE COLOUR OF LIGHT:** The refractive index of a transparent medium decreases with the increase of wavelength of light (it is most for the violet light and least for the red light), therefore the critical angle for a pair of media is least for the violet light and most for the red light i.e. the critical angle increases with the increase in wavelength of light.

(2) **DEPENDENCE ON TEMPERATURE:** On increasing the temperature of medium, its refractive index decreases, so the critical angle for that pair of media increases. Thus critical angle increases with an increase in temperature.

T.I.R.



TOTAL INTERNAL REFLECTION

When a ray of light travelling in a denser medium, is incident at the surface of a rarer medium at the angle of incidence greater than the critical angle for the pair of media, the ray is totally reflected back into the denser medium. This phenomenon is called total internal reflection.

Essential conditions for the total internal reflection

(1) The light must travel from a denser to a rarer medium.

(2) The angle of incidence must be greater than the critical angle for the pair of media.

Glass (D) ⟶ Air (R)

$$\sin i_c = \frac{1}{a\mu_g}$$

$$= \frac{1}{\left(\frac{3}{2}\right)}$$

$$\sin i_c = \frac{2}{3}$$

$$\boxed{i_c = \sin^{-1}\left(\frac{2}{3}\right) \approx 42^\circ}$$

Diamond (D) ⟶ Air (R)

$$\sin i_c = \frac{1}{a\mu_d}$$

$$= \frac{1}{2.42}$$

$$\sin i_c = 0.413$$

$$\boxed{i_c = \sin^{-1}(0.413) \approx 24.4^\circ}$$



HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

- A water pond appears to be 2.7 m deep. If the refractive index of water is 4/3, find the actual depth of the pond.

[Ans. 3.6 m]

- A coin is placed at the bottom of a beaker containing water (refractive index = 4/3) to a depth of 12 cm . By what height the coin appears to be raised when seen from vertically above?

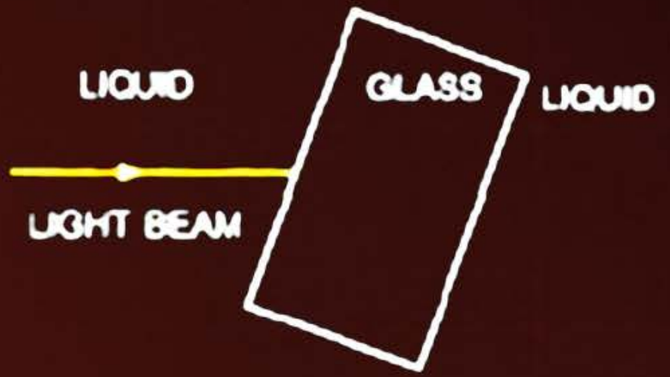
[Ans. 3 cm]

- A postage stamp kept below a rectangular glass slab of refractive index 1.5 when viewed from vertically above it, appears to be raised by 7.0 mm . Calculate the thickness of the glass slab.

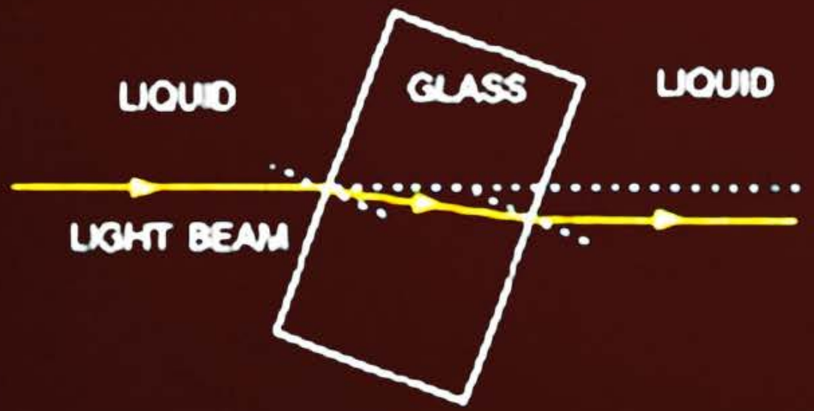
[Ans. 2.1 cm]

QUESTIONS

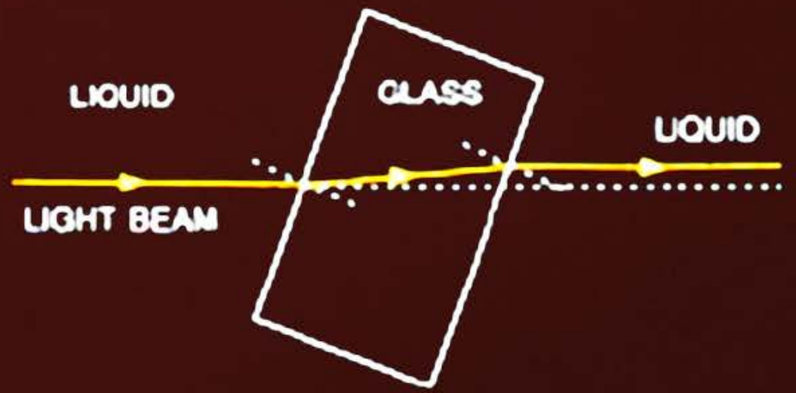
The diagram below shows a glass block suspended in a liquid. A beam of light of single colour is incident from liquid on one side of the block.



- Draw diagrams to show how does the light bend when it travels from liquid to glass and then to liquid if (i) the light slows down in glass, and (ii) the light speeds up in glass.



light slows down in glass



Thank You